

◆ Slide 1 — Title Slide

Proactive Carbon Intelligence:
ML-Based Emission Monitoring and Anomaly Detection

Supervised By: [Faculty Name]

Group No: 13

Aryan Sharma – 2241019403
Anand Kumar Sharma – 2241019398
Saswat Dash – 2241004182
Suyash Kumar Anand – 2241001074

Department of CSE
ITER, SOA University

◆ Slide 2 — Presentation Outline

- Problem Statement & Motivation
 - Literature Review
 - Research Gap
 - Proposed Solution & Architecture
 - Algorithms Used
 - Dataset & Tools
 - Work Done So Far
 - Preliminary Results
 - Future Work Plan
-

◆ Slide 3 — Problem Statement

Industrial systems generate high carbon emissions due to inefficient energy usage.

Existing monitoring systems are reactive and detect issues only after they occur.

There is a need for a predictive system to:

- Estimate emissions
 - Detect abnormal behavior
 - Improve sustainability
-

◆ Slide 4 — Motivation

- Increasing global concern over carbon emissions

- Need for sustainable industrial operations
 - Lack of intelligent monitoring systems
 - Opportunity to apply machine learning
-

◆ Slide 5 — Literature Review

- Machine learning is widely used for energy prediction
 - Time-series models are used for forecasting
 - Isolation Forest is used for anomaly detection
 - Ensemble models improve prediction accuracy
-

◆ Slide 6 — Research Gap & Improvements

- Most systems are reactive, not predictive
- Limited integration of prediction and anomaly detection
- Lack of user-friendly monitoring systems

Improvement:

- Develop a predictive and integrated monitoring system
-

◆ Slide 7 — Proposed Solution & Architecture

- Predict power output using environmental data
 - Estimate CO₂ emissions using emission factor
 - Detect anomalies using machine learning
 - Provide insights through visualization
-

◆ Slide 8 — Workflow Diagram (IMPORTANT)

👉 Draw diagram in PPT and add this text:

Environmental Data (AT, V, AP, RH)



Power Prediction Model



CO₂ Estimation



Residual Calculation



Anomaly Detection (Isolation Forest)

◆ Slide 9 — Proposed Architecture (cont.)

Modules:

- Power Prediction Module
 - Emission Estimation Module
 - Anomaly Detection Module
 - Visualization Module (Planned)
-

◆ Slide 10 — Dataset & Tools

Dataset:

Combined Cycle Power Plant Dataset

Features:

- Temperature (AT)
- Vacuum (V)
- Pressure (AP)
- Humidity (RH)
- Power Output (PE)

CO₂ derived using:

$$\text{CO}_2 = \text{PE} \times 0.82$$

Tools:

- Python
 - Pandas, NumPy
 - Scikit-learn
 - Matplotlib
-

◆ Slide 11 — Progress in Implementation

- Data preprocessing completed
 - CO₂ emission derived from power output
 - Linear Regression model implemented
 - Gradient Boosting model implemented
 - Model comparison performed
 - Anomaly detection using Isolation Forest
 - Visualization of anomalies completed
-

◆ Slide 12 — Preliminary Results

Linear Regression RMSE $\approx$ 4.47

Gradient Boosting RMSE $\approx$ 3.80 (better performance)

Anomaly Detection:

- 192 anomalies detected
 - Approx. 2% abnormal behavior identified
-

◆ Slide 13 — Future Work Plan

- Implement optimization module
 - Develop Streamlit dashboard
 - Improve model performance
 - Integrate real-time monitoring
-

◆ Slide 14 — References

- UCI Machine Learning Repository
- Scikit-learn Documentation
- Research papers on emission prediction
- Research papers on anomaly detection